FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – II

(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	31.	**C**	61.	**B**
2.	**C**	32.	**B**	62.	**C**
3.	**A**	33.	**B**	63.	**A**
4.	**A**	34.	**C**	64.	**A**
5.	**B**	35.	**B**	65.	**A**
6.	**B**	36.	**B**	66.	**A**
7.	**B**	37.	**C**	67.	**A**
8.	**D**	38.	**B**	68.	**C**
9.	**A**	39.	**B**	69.	**C**
10.	**C**	40.	**C**	70.	**B**
11.	**A**	41.	**D**	71.	**B**
12.	**D**	42.	**B**	72.	**A**
13.	**C**	43.	**C**	73.	**C**
14.	**B**	44.	**B**	74.	**D**
15.	**B**	45.	**B**	75.	**A**
16.	**C**	46.	**C**	76.	**D**
17.	**C**	47.	**C**	77.	**B**
18.	**C**	48.	**B**	78.	**D**
19.	**B**	49.	**C**	79.	**A**
20.	**B**	50.	**B**	80.	**D**
21.	**A**	51.	**A**	81.	**A**
22.	**D**	52.	**B**	82.	**A**
23.	**D**	53.	**D**	83.	**A**
24.	**A**	54.	**D**	84.	**B**
25.	**B**	55.	**B**	85.	**D**
26.	**B**	56.	**B**	86.	**A**
27.	**C**	57.	**C**	87.	**B**
28.	**C**	58.	**A**	88.	**A**
29.	**C**	59.	**B**	89.	**C**
30.	**C**	60.	**A**	90.	**A**

Physics

PART – I

SECTION – A

1 C

Sol. $m\frac{dv}{dt} = -6\pi\eta rv$

$(1)\frac{dv}{dt} = 6\pi\left(\frac{1}{18\pi}\right)(1)v$

$\frac{dv}{dt} = -\frac{v}{3}$

$\frac{dv}{v} = -\frac{dt}{3}$

$[\ln v]_2^{0.5} = \left[-\frac{t}{3}\right]_{t_1}^{t_2}$

$\ln 4 = \frac{\Delta t}{3}$

$\Delta t = 3 \ln 4$

2 C

Sol. $\frac{\lambda_R}{\lambda_S} = 1.5$

So, the rate of disintegration of R will be 1.5 times that of S. Thus, the half-life of S will be 1.5 times that of R. So, two half lives of S will be equal to the three half –lives of R.

$\frac{N_R}{N_S} = \frac{0.25}{0.25} = 1$

3. A

Sol. From perpendicular axis theorem $I_z = I_x + I_y = 2I$

4. A

Sol. Since collision is elastic $\frac{1}{2}mv^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2$

$\Rightarrow v_1^2 + v_2^2 = v^2$ (i)

and momentum conservation given $m\vec{v} = m\vec{v}_1 + m\vec{v}_2$

$\Rightarrow \quad \vec{v}_1 + \vec{v}_2 = \vec{v}$ (ii)

Equation (i) and (ii) shows that angle between the velocity of both bodes after collision is 90^0 thus angle made by another body = 90 – 15 = 75^0

5. B

Sol. $T = 2\pi\sqrt{\frac{\ell}{g_{ef}}}$, $g_{ef} = \left(\frac{mg + qE}{m}\right)$

6. B

Sol. $muy = mVy + \frac{3}{2}MR^2\omega$ where u is speed of ball before collision, V speed of ball after collision and ω is angular speed of cylinder after collision.

$mu = mV + MR\omega$ (COM)

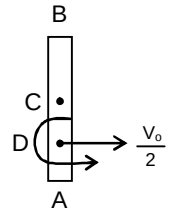
$\therefore\ y = \frac{3R}{2}$.

7. B

Sol. $d = \frac{mv_0}{qB} + \frac{2mv_0}{2qB}$

8. D

Sol. By COAM about point D (ie about new centre of mass)



$$MV_o\frac{L}{4} = \left[\frac{ML^2}{12} + \frac{ML^2}{16} + \frac{ML^2}{16}\right]\omega$$

or, $w = \frac{6v_o}{5L}$

Now, $t = \frac{\theta}{w} = \frac{(\pi/2)}{w} = \frac{5\pi L}{12v_o}$

9. A

Sol. $x^2 + y^2 = \ell^2$

$$\frac{dy}{dt} = -\frac{x}{y}\left(\frac{dx}{dt}\right)$$

If $\theta = 37°$

$$\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{4}{3}v_o$$

$$\vec{v}_{cm} = \frac{mv_o\hat{i} - m\frac{4v_o}{3}\hat{y}}{2m}$$

$$\left|\vec{V}_{cm}\right| = \frac{5v_o}{6}$$

10. C

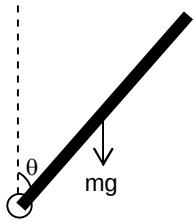
Sol. $4\times\ell\frac{12}{24} = \frac{\varepsilon_1 r_2 - r_1\varepsilon_2}{r_1 + r_2}$

11. A

Sol. The figure shows the rod at an angle θ to the vertical. If we take torques about the pivot we need not be concerned with the force due to the pivot. The torque due to the weight is $1/2mgL\sin\theta$, so the second law for the rotational motion is

$\frac{mgL}{2}\sin\theta = \frac{ML^2}{3}\alpha$

Thus $\alpha = \frac{3g\sin\theta}{2L}$



When the rod is horizontal $\theta = \pi/2$ and $\alpha = 3g/2L$.

The tangential linear acceleration of the free end is $a_t = \alpha L = \frac{3g}{2}$

12. D

Sol. Moment of inertia of wheel about its axis,

$I = \frac{mR^2}{2} = \frac{2\times(0.1)^2}{2} = 0.01$ kg-m^2

Now angular displacement, $\Delta\theta = 2\pi n = 2\pi\times5 = 10\ \pi$ rad

$\therefore$ Angular retardation required, $\alpha = \frac{\omega^2}{2\Delta\theta} = \frac{(2\pi)^2}{2\times10\pi} = \pi/5$ rad/s^2

$\therefore$ Torque required $\tau = I\alpha \Rightarrow FR = I\alpha$

$\therefore F = \frac{I\alpha}{R} = \frac{0.01}{0.10}\times\frac{\pi}{5} = 2\pi\times10^{-2}$ N

13. C

Sol. Energy released from n = 5 to n = 1

$$E = 13.6\left(\frac{1}{1^2}-\frac{1}{5^2}\right) = 13.6\times\frac{24}{25}\ \text{eV}$$

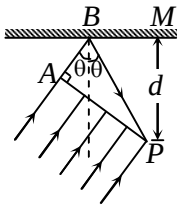
Momentum of photon = $13.6\times\frac{24}{25}\times\frac{1.6\times10^{-19}}{3\times10^8}$

Momentum of H-atom = $\frac{13.6\times24\times1.6\times10^{-19}}{25\times3\times10^8}$

Velocity of H-atom = $\frac{13.6\times24\times1.6\times10^{-19}}{75\times10^8\times1.67\times10^{-27}} =$ 4 m/s

14. B

Sol. $BP = \frac{MP}{\cos\theta}$



$AB = BP\cos2\theta = \frac{MP}{\cos\theta}\cos2\theta$

$AB + BP = n\lambda + \frac{\lambda}{2}$

$\frac{d\cos2\theta}{\cos\theta} + \frac{d}{\cos\theta} = \frac{d}{\cos\theta}(2\cos^2\theta - 1 + 1) = 2d\cos\theta = n\lambda + \frac{\lambda}{2}$ for n = 1, $\cos\theta = \frac{3\lambda}{4d}$

15. B

Sol. As accelerating voltage is increased, energy increases; the λ_{min} decreases and X-rays are getting harder (less wavelength) and penetrating power increases if filament current increases, more electrons are emitted.

16. C

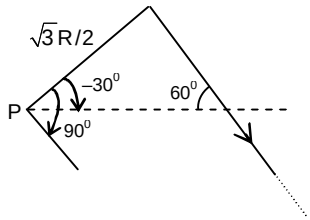
Sol. $\mu = \frac{3}{2}$; $V = 8$; $m = \frac{1}{3}$

$1 + m = \frac{V}{f} = \frac{V}{2R}$

17. C

Sol. $B = \frac{\mu_0 I}{4\pi\sqrt{3}\,R/2}\left[\sin 90^o + \sin(-30^o)\right]$

$= \frac{\mu_0 I}{4\sqrt{3}\,R}$



18. C

Sol. Mass per unit length, $\mu = \frac{m}{\ell} = \frac{\rho A \ell}{\ell} = \rho A$

$\mu_s = \mu_{A\ell} = 78 \times 10^{-4}$ kg/m

$\therefore$ Speed of wave is same in both wire

$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80\times10^4}{78}} = \frac{2\times10^2}{\sqrt{3.9}}$

$\nu_{min} = \frac{V}{\lambda_{max}} = \frac{200}{\sqrt{3.9}\times0.3}$ $\left[\frac{\lambda_{max}}{2} = 15 \text{ cm for C as a node}\right] = 337.5$ Hz

19. B

Sol. $100°C - Ir - \frac{I}{2}r - Ir = 0°C$

$\Rightarrow$ $Ir = 40°C$; $\therefore$ $t_F = 0° + Ir = 40°C$

20. B

Sol. $V = \frac{0.5\times3}{(0.5+1)} = 1\,m/s \Rightarrow K_{max} = U_{max}$

$\frac{1}{2}\times1.5\times1^2 = \frac{1}{2}\times600\times A^2$

A= 5 cm

$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{600}{1.5}} = \frac{10}{\pi}$

$f = \frac{10}{\pi}$ Hz

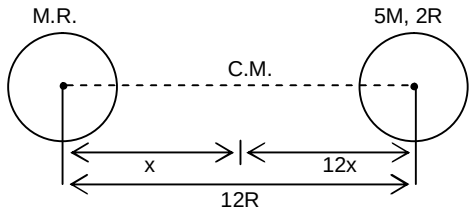
21. A

Sol. $H = -k4\pi r^2\frac{dT}{dr} \Rightarrow H \propto \frac{r_1 r_2}{r_2 - r_1}$

22. D

Sol. Initially

$Mx = 5M(12-x) \Rightarrow x = 10R$

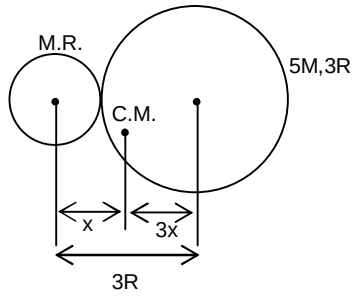


Finally

$Mx = 5M(3R - x)$

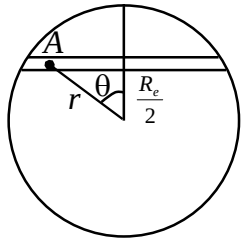
$x = 2.5R$

Distance travelled by smaller block = 10R – 2.5R = 7.5 R



23. D

Sol. Let mass of the earth is M_e and mass of object is m.



Force on object at $A = \left(\frac{GM_em}{R^3}r\right)$ towards centre.

Force on object along the tunnel $= \left(\frac{GM_em}{R^3}r\right)\frac{\sqrt{r^2 - \frac{R^2}{4}}}{r}$

Acceleration of object along the tunnel $= \left(\frac{GM_e}{R^3}\right)\sqrt{r^2 - \frac{R^2}{4}}$

24. A

Sol. $\psi_1 = A\cos(\omega t - kx)$ …(i)

$\psi_2 = A\cos(\omega t - ky)$ …(ii)

Put $y = x + \frac{(2n+1)}{2}\lambda$ in (ii)

$\psi_2 = A\cos\left[\omega t - k\left(x + (2n+1)\frac{\lambda}{2}\right)\right]$

$\therefore \quad \psi_2 = A\cos(\omega t - kx - (2n+1)\pi)$

$\therefore \quad \psi_2 = -A\cos(\omega t - kx) = -\psi_1$

$\therefore$ Shape of path is straight line.

25. B

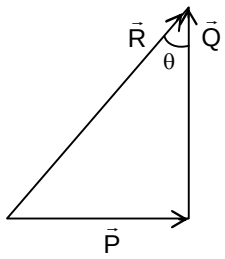
Sol. The charge q is completely closed by four identical given box. Therefore, the required flux = $\frac{1}{4}\frac{q}{\epsilon_o}$.

26. B

Sol. Required value = $(V_C \cdot i_C)_{max} = \frac{\xi}{2}\cdot\frac{\xi}{2R} = \frac{\xi^2}{4R}$

27. C

Sol. $\cos\theta = \dfrac{Q}{R}$



28. C

Sol. Let u be the initial speed of the particle

Then $v^2 = u^2 - 2gh$

or $u^2 = v^2 + 2gh$

or $u_x^2 + u_y^2 = v_x^2 + v_y^2 + 2gh \qquad (v_x = u_x)$

or $u_y^2 = v_y^2 + 2gh$

or $u_y^2 = (2)^2 + (2)(10)(0.4) = 12$

$\therefore \quad u_y = \sqrt{12}$ or $2\sqrt{3}$ m / s

and $u_x = v_y = 6$ m/s

$\therefore \quad \tan\theta = \dfrac{u_y}{u_x} = \dfrac{2\sqrt{3}}{6} = \dfrac{1}{\sqrt{3}}$

or $\theta = 30°$

29. C

Sol. Power $P = F.v = \dfrac{K}{v}.v = K = \text{constant}$

$\therefore \; W = P.t = Kt$

30. C

Sol. Liquid exerts force B on block in upward direction and on beaker in downward direction.

Chemistry PART – II

SECTION – A

31. C

Sol. Mass of 1.8 mL water = $1.8 \times 1 = 1.8$ g

No. of moles = $\frac{1.8}{18} = 0.1$

One molecule of H_2O contains 10 electrons

$\therefore$ No. of electrons in 0.1 mole = $0.1 \times 10 = 1$ mole = 6.023×10^{23}

32. B

Sol. $r_n = n^2 a_0$ for H – atom

33. B

Sol. The N atom in NO_2^+ undergoes sp-hybridization with bond angle 180°.

34. C

Sol. $KI_3 \longrightarrow K^+ + I_3^-$

$\therefore$ The oxidation number of iodine in I_3^- is $-\frac{1}{3}$

35. B

Sol. $Cu \longrightarrow Cu^{2+} + 2e^-$ $\quad \Delta G_1^0$

$2Cu^{2+} + 2e^- \longrightarrow 2Cu^+$ $\quad \Delta G_2^0$

$\Delta G_f^0 = \Delta G_1 + \Delta G_2$

$\Delta G_f^0 = -2.303 RT \log K$

36. B

Sol. At high pressure 'a' is neglected due to repulsion between gas molecules.

$\therefore$ P(V – b) = RT or PV = RT + Pb

37. C

Sol. Milliequivalent of H_2SO_4 = 25.1 x 0.02 = 0.502

Equivalent = 0.502 x 10^{-3}

Eq of H_2SO_4 = Eq of $CaCO_3$

$\therefore$ Mass of $CaCO_3$ = Eq x 50 = 0.502 x 10^{-3} x 50 = 0.0251

100 mL H_2O = 100 g contains 0.0251 g $CaCO_3$

10^6 g of water will contain $\frac{0.0251}{10^2} \times 10^6 = 251$ g of $CaCO_3$

$\therefore$ The degree of hardness = 251 ppm

38. B

Sol. $\frac{\frac{n_{H_2}}{10}}{\frac{n_{O_2}}{10}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}}$

$$\text{or, } \frac{2/2}{10}\times\frac{10}{n_{O_2}}=\sqrt{\frac{32}{2}}=4$$

$$\therefore \frac{1}{n_{O_2}}=4\Rightarrow n_{O_2}=\frac{1}{4}\Rightarrow W_{O_2}=\frac{1}{4}\times 32=8\,g$$

39. B

Sol. Phosphorus contains maximum positive charge in $PClF_4$

40. C

Sol. The value of K_C does not change either by adding or removing reactants or products from the equilibrium mixture.

41. D

Sol. The products are due to coupling of different alkyl groups. If $CH_3CH_2CH_2Cl$ is RCl and $CH_3\underset{\underset{Cl}{|}}{C}HCH_3$ is R'Cl, the products will be RR, R'R' and RR'.

42. B

Sol. For first order reaction, the half-life time is given as: $t_{1/2}=\frac{0.693}{K}$

$\therefore$ The half-life is independent of the concentration of reactant.

43. C

Sol. K_{sp} changes only by changing temperature.

44. B

Sol. Heat of hydrogenation of the least substituted alkene is maximum.

45. B

Sol. The diene is $CH_3CH = CH - CH_2 - CH = CHCH_3$

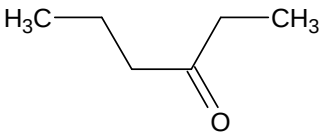
46. C

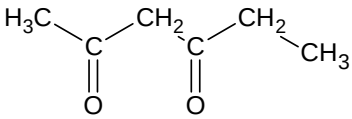
Sol. $W = -2.303n\,RT\log\frac{V_2}{V_1}=-2.303\times 1.2\times 8.314\times 10^{-3}\times 300\log\frac{10V_1}{V_1}$ = -6.892 kJ mol^{-1}

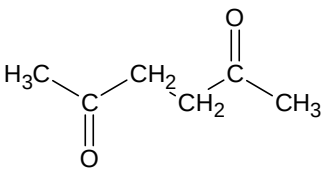
47. C

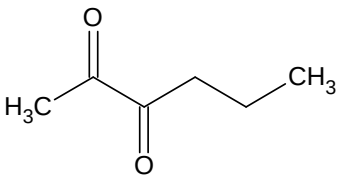
Sol. Breaking of C – O bond results in formation of most stable tertiary carbocation.

48. B

Sol. (A) 

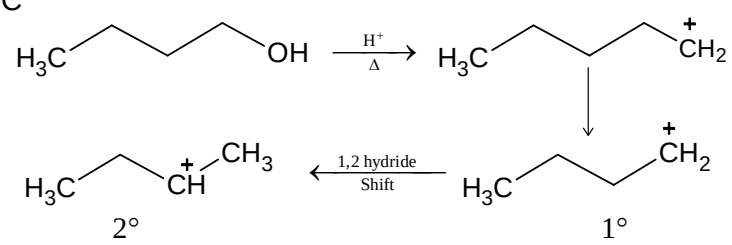
(B) 

(C) 

(D) 

49. C

Sol.



50. B

Sol. S^{2-} makes the CCP lattice where Zn^{2+} ions are present at the alternate tetrahedral voids.

51. A

Sol. Using

$\Delta U = nC_V \times \Delta T$

$\Delta T = 100$ n = 3.5

$C_V = 2 \times 10^{-2}$

$\Delta U = 7$

52. B

Sol. $4BCl_3 + 6H_2 + C \longrightarrow B_4C + 12HCl$

The reaction takes place in the preparation of bullet proof fabric B_4C.

53. D

Sol. This is due to lanthanoid contraction on 6th period.

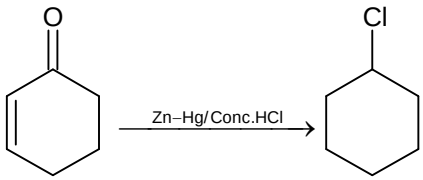
54. D

Sol. $C_6H_5CHO + CH_3COOCOCH_3 \xrightarrow[\Delta]{CH_3COONa} C_6H_5 - CH = CH - COOH + CH_3COOH$

$CO_2 + C_6H_5 - CH = CH_2 \xleftarrow{\Delta}$

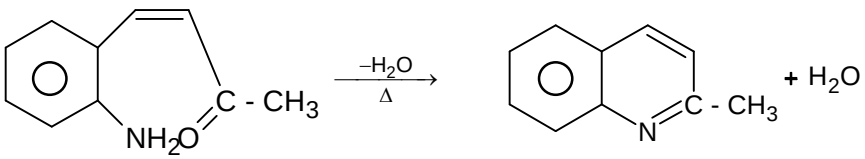
55. B

Sol.



56. B

Sol.



57. C

Sol. Octahedral complex with C.N number 6.

58. A

Sol. No free Cl^- in the ionisation sphere.

59. B

Sol. $SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$

$SiO_2 + Na_2CO_3 \longrightarrow Na_2SiO_3 + CO_2$

60. A

Sol. Having two possible donating site but donate from one only.

Mathematics PART – III

SECTION – A

61. B

Sol. $x^{(\log_2 x)^2 - 6\log_2 x + 12} = 256$;

$\left(t^2 - 6t + 12\right)t = 8$ where $t = \log_2 x$;

$\Rightarrow (t-2)^3 = 0$

$\Rightarrow t = 2 \Rightarrow x = 4$

62. C

Sol. Note that $x \in [-2, 2]$ for x to be real.

Let $x = 2\cos\theta$ where $\theta \in [0, \pi]$.

[Note : $x + 2 \geq 0$ and $2 - \sqrt{2+x} \geq 0 \Rightarrow x \in [-2,2]$]

$$x = \sqrt{2 + \sqrt{2 - 2\cos\frac{\theta}{2}}}$$

$$2\cos\theta = \sqrt{2 + \sqrt{2 - 2\cos\frac{\theta}{2}}}$$

$$= \sqrt{2 + \sqrt{2\left(1 - \cos\frac{\theta}{2}\right)}} = \sqrt{2 + 2\sin\frac{\theta}{4}}$$

$$= \sqrt{2 + 2\cos\left(\frac{\pi}{2} - \frac{\theta}{4}\right)}$$

$$= \sqrt{2\left(1 + \cos\left(\frac{\pi}{2} - \frac{\theta}{4}\right)\right)}$$

$$= \sqrt{4\cos^2\left(\frac{\pi}{4} - \frac{\theta}{8}\right)}$$

$$2\cos\theta = 2\cos\left(\frac{\pi}{4} - \frac{\theta}{8}\right);$$

$$\therefore \quad \theta = \frac{\pi}{4} - \frac{\theta}{8}$$

$$\Rightarrow \quad \frac{9\theta}{8} = \frac{\pi}{4} \Rightarrow \quad \theta = \frac{2\pi}{9}$$

Hence $x = 2\cos\frac{2\pi}{9} = 2\cos 40^o$

63. A

Sol. $$\frac{\left(\alpha^5 + \beta^5\right)}{\left(\alpha^3 + \beta^3\right)} = \frac{11}{3}$$

$$\frac{1}{\left(\alpha^2+\beta^2-\alpha\beta\right)}\left(\alpha^4+\beta^4-\alpha\beta\left(\alpha^2+\beta^2-\alpha\beta\right)\right)=\frac{11}{3}$$

$$\frac{\left(\alpha^4+\beta^4\right)}{\left(\alpha^2+\beta^2-\alpha\beta\right)}-\alpha\beta=\frac{11}{3}$$

$$\frac{25-2(\alpha\beta)^2}{5-\alpha\beta}-\alpha\beta=\frac{11}{3}$$

Let $\alpha\beta=t$; by coefficient $\alpha\beta=2$

64. A

Sol. $T_1=\frac{1}{4}(4-3)=1-\frac{3}{4}$

$$T_2=\frac{1(3)}{4(6)}[6-5]=\frac{3}{4}-\frac{1(3)(5)}{4(6)}$$

$$T_3=\frac{1(3)(5)}{4(6)(8)}[8-7]=\frac{1(3)(5)}{4(6)}-\frac{1(3)(5)(7)}{4(6)(8)}$$

$$\vdots$$

$$\therefore T_1+T_2+T_3.....=1$$

65. A

Sol. We have $\frac{P}{Q^2}=\frac{\left(x-\frac{1}{x}\right)\left(x^2+\frac{1}{x^2}+1\right)}{\left(x-\frac{1}{x}\right)^2}$

As $x\in(1,\infty)$, so $\left(x-\frac{1}{x}\right)>0$

Now $\frac{P}{Q^2}=\frac{\left(x-\frac{1}{x}\right)^2+3}{\left(x-\frac{1}{x}\right)}=\left(x-\frac{1}{x}\right)+\frac{3}{\left(x-\frac{1}{x}\right)}$

$$\Rightarrow \frac{P}{Q^2}\geq 2\sqrt{\left(x-\frac{1}{x}\right)\left(\frac{3}{x-\frac{1}{x}}\right)}=2\sqrt{3}$$ (Using A.M. $\geq$ G.M.)

66. A

Sol. $\sin\left(\frac{\pi}{3}+x\right)+\sin(\alpha+x)\geq 0$

$$2\sin\left(\frac{\pi}{6}+\frac{\alpha}{2}+x\right)\cos\left(\frac{\pi}{6}-\frac{\alpha}{2}\right)\geq 0 \qquad \ldots\ldots\ldots(i)$$

Now, $\sin\left(\frac{\pi}{6}+\frac{\alpha}{2}+x\right)\in[-1,1]$ for $x\in R$

Hence equation (i) will be true

$\forall\, x\in R$ if $\frac{\pi}{6}-\frac{\alpha}{2}=(2k+1)\frac{\pi}{2}$

Possible $k=-1$

$\therefore \quad \frac{\alpha}{2}=\frac{\pi}{6}+\frac{\pi}{2}=\frac{2\pi}{3}$

$\Rightarrow \quad \alpha=\frac{4\pi}{3}\equiv\frac{p\pi}{q}$

$\Rightarrow \quad (p+q)=7$

67. A

Sol. $$\sum_{R=1}^{N}\frac{1}{N^2+R^2}\sum_{K=1}^{MR}\frac{N^2}{N^2+\left(\frac{K}{M}\right)^2}$$

$$\lim_{N\to\infty}\lim_{M\to\infty}\sum_{R=1}^{N}\frac{1}{N^2+R^2}\int_0^2\frac{N^2}{N^2\times x^2}dx$$

$$\lim_{N\to\infty}\sum_{R=1}^{N}\frac{N\tan^{-1}\left(\frac{R}{N}\right)}{N^2+R^2}=\frac{\pi^2}{32}$$

68. C

Sol. We must have $m<0$

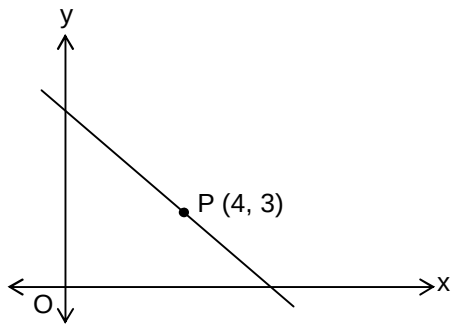
(If $m>0$ then either x or y intercept must be negative)

Let $(y-3)=m(x-4)$

Now, x – intercept $=4-\frac{3}{m}$

and y – intercept = 3 – 4m

For x intercept to be positive prime and y – intercept to be positive integer, possible values of m are i.e. m = –1 or m = –3.



69. C

Sol. Given equation of lines

$\sqrt{3}y^2-4xy+\sqrt{3}x^2=0$

$\Rightarrow \quad \sqrt{3}x^2-3xy-xy+\sqrt{3}x^2=0$

$\Rightarrow \quad \left(\sqrt{3}y-x\right)\left(y-\sqrt{3}x\right)=0$

$\Rightarrow \quad y = \frac{x}{\sqrt{3}}, y = \sqrt{3}x$

$\therefore \quad \angle APO = 75^o$

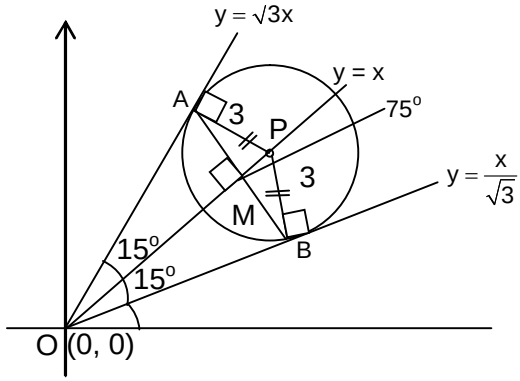
In ΔAMP, $\sin 75^o = \frac{AM}{3}$

$\Rightarrow \quad AM = 3\sin 75^o$

Now, length of chord of contact $AB = 2AM$

$= 2\left(3\sin 75^o\right) = 6\sin 75^o$

$= 6\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) = 3\frac{\left(\sqrt{3}+1\right)}{\sqrt{2}}$



70. B

Sol. (a) It should be non – singular

(c) since $A^{-1} = \frac{\text{adj}\,A}{|A|}$, hence $\text{adj}\,A$ must be a non singular matrix. Its inverse must exist.

(d) It should be $F(x+y)$

71. B

Sol. Case – 1 : abbc

We can choose a in 9 ways, b in 9 ways, and c in 9 ways. Hence, there are 9^3 possibilities for this case.

Case – 2 : bbac

There are 9 possibilities for b, and 9 possibilities for each a and c. Once again, we have that there are 9^3 possibilities for this case.

Case – 3 : acbb

Once again, there are 9 possibilities for a and c, and 9 possibilities for b. Hence, there are 9^3 possibilities for this case.

Case – 4 : aabb

We have 9 possibilities for a and 9 possibilities for b.

We thus have 81 possibilities for this case.

We can now sum up our answers, to get a total of 3.

$9^3 + 81 = 2268$ total ways.

72. A

Sol. Any element, is of the form of $\frac{8!}{t_1!t_2!t_3!t_4!}\left(x^2\right)^{t_1}\left(\frac{1}{x^2}\right)^{t_2}(y)^{t_3}\left(\frac{1}{y}\right)^{t_4}$ where

$t_1 + t_2 + t_3 + t_4 = 8;\ t_1 \geq 0$

The constant term occur when $t_1 = t_2$ and $t_3 = t_4$

So, $t_1 + t_3 = 4 \Rightarrow (0,4),(4,0);(1,3),(3,1);(2,2)$

$\Rightarrow$ Constant term:

$\frac{8!}{0!0!4!4!} + \frac{8!}{4!4!0!0!} + \frac{8!}{1!1!3!3!} + \frac{8!}{3!3!1!1!} + \frac{8!}{2!2!2!2!}$

$= (2\times 70) + (2\times 1120) + 2520 = 4900$

73. C

Sol. Coefficient of x^{11} in the expansion of $(1+x)^5(3+x)^4(7+x)^3$

$=\left({}^5C_0+{}^5C_1.x+.....+{}^5C_4.x^4+{}^5C_5.x^5\right)$

$\left({}^4C_0.3^4+{}^4C_1.3^3x+.....+{}^4C_3.3x^3+{}^4C_4.x^4\right)$

$\left({}^3C_0.7^3+{}^3C_1.7^2x+{}^3C_2.7x^2+{}^3C_3+.x^3\right)$

$={}^5C_4.{}^4C_4.{}^3C_3+\left({}^4C_3.3\right){}^5C_5.{}^3C_3+\left({}^3C_2.7\right){}^5C_5.{}^4C_4$

$=5.1.1+4.3.1.1+3.7.1.1=38$

74. D

Sol. $f(x)=\frac{1}{2}\ln\left(\sqrt{\sqrt{x^2+1}+x}\pm\sqrt{\sqrt{x^2+1}-x}\right)^2$

$=\frac{1}{2}\ln\left\{\sqrt{x^2+1}+x+\sqrt{x^2+1}-x\pm2\sqrt{\sqrt{\left(x^2+1\right)^2}-x^2}\right\}$

$=\frac{1}{2}\ln\left\{2\sqrt{x^2+1}\pm2\right\}$

An even function hence neither one – one nor onto.

75. A

Sol. Denominator $\neq 0$ and numerator

$\left(x^4+3x^2+2\right)-2x\left(x^2+1\right)=\left(x^2+1\right)\left((x-1)^2+1\right)$

76. D

Sol. As, $T_n=\cot^{-1}\left[\frac{(n+1)(n+2)}{2}x+\frac{2}{x}\right]$

$\Rightarrow T_n=\tan^{-1}\left(\frac{2x}{(n+2)(n+1)x^2+4}\right)$

$\therefore T_n=\tan^{-1}\left(\left(\frac{n+2}{2}\right)x\right)-\tan^{-1}\left(\left(\frac{n+1}{2}\right)x\right)$

So, $S_n=\tan^{-1}\left(\left(\frac{n+2}{2}\right)x\right)-\tan^{-1}x$

$\Rightarrow \lim_{n\to\infty}S_n=\frac{\pi}{2}-\tan^{-1}x$

$=\cot^{-1}x=1$ (given)

$\Rightarrow \quad x=\cot 1$

77. B

Sol. $n<\sqrt{n^2+n+1}<n+1$

Hence $\left[\sqrt{n^2+n+1}\right]=n$

$\therefore I=\lim_{n\to\infty}\left(\sqrt{n^2+n+1}-n\right)=\frac{1}{2}$

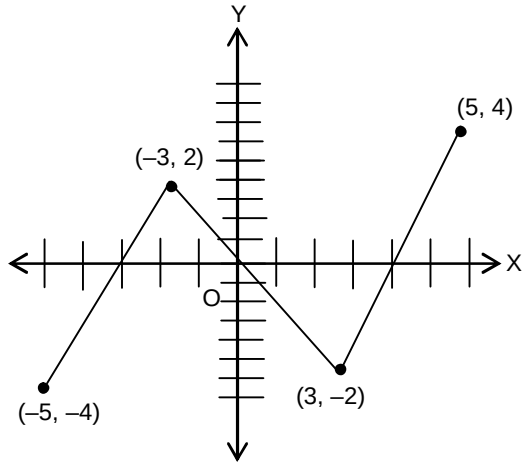
78. D

Sol. $f(x)+f(-x)=0$

$\Rightarrow f(x)$ is an odd function. Since points (–3, 2) and (5, 4) lie on the curve.

$\therefore \quad (3,-2)$ and $(-5,-4)$ will also lie on the curve

For minimum number of roots, graph of continuous function $f(x)$ is as follows.



From the above graphs of $f(x)$ it is clear that equation $f(x)=0$ has at least three real roots.

79. A

Sol. $\dfrac{f(h)-f(0)}{h}+\dfrac{2\left[f(-2h)-f(0)\right]}{-2h}=f'\left(0^{+}\right)+2f'\left(0^{-}\right)$

$=3f'(0)$

80. D

Sol. $f'(x)=\begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$f''(x)=\begin{vmatrix} 6x & -\sin x & -\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$f'''(x)=\begin{vmatrix} 6 & -\cos x & \sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$\Rightarrow f'''(0)=\begin{vmatrix} 6 & -1 & 0 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}=0$ (two identical rows)

81. A

Sol. Differentiate the given integral with respect to x

$$u\frac{d^2v}{dx^2} = u\frac{d^2v}{dx^2} + \frac{du}{dx}.\frac{dv}{dx} - v\frac{d^2u}{dx^2} - \frac{dv}{dx}.\frac{du}{dx} + \frac{dw}{dx}$$

$$\Rightarrow \frac{dw}{dx} = v\frac{d^2u}{dx^2}$$

$$\therefore\ dw = v\frac{d^2u}{dx^2}dx$$

$$\therefore\ w = \int v\frac{d^2u}{dx^2}dx$$

82. A

Sol. $$L = \lim_{n\to\infty}\frac{n}{(n!)^{1/n}} = \lim_{n\to\infty}\frac{n}{(1.2.3.....n)^{1/n}}$$

$$= \lim_{n\to\infty}\frac{1}{\left(\frac{1}{n}.\frac{2}{n}.\frac{3}{n}...\frac{n}{n}\right)^{1/n}}$$

$$\ln\ L = -\frac{1}{n}\left[\ln\left(\frac{1}{n}\right) + \ln\left(\frac{2}{n}\right) + \ln\left(\frac{3}{n}\right) + ... + \ln\left(\frac{n}{n}\right)\right]$$

General term of ln L is

$$T_r = -\frac{1}{n}\ln\frac{r}{n}$$

$$\therefore S = \lim_{n\to\infty} -\frac{1}{n}\sum_{r=1}^{n}\ln\frac{r}{n} = -\int_0^1 \ln x\,dx = -\left[x\ln x - x\right]_0^1$$

$$= \left[(0-1)-(0)\right] = 1$$

$\therefore$ Hence ln L = 1

L = e

83. A

Sol. $$I = \int_0^1 x\ln\left(\frac{x+2}{2}\right)dx = \int_0^1 x\left(\ln(x+2) - \ln 2\right)dx$$

$$\therefore\ I = \int_0^1 x\ln(x+2)dx - \ln 2\int_0^1 x\,dx$$

$$\text{Hence } I = \left[\ln(x+2).\frac{x^2}{2}\right]_0^1 - \int_0^1 \frac{x^2}{x+2}dx - \frac{\ln 2}{2}$$

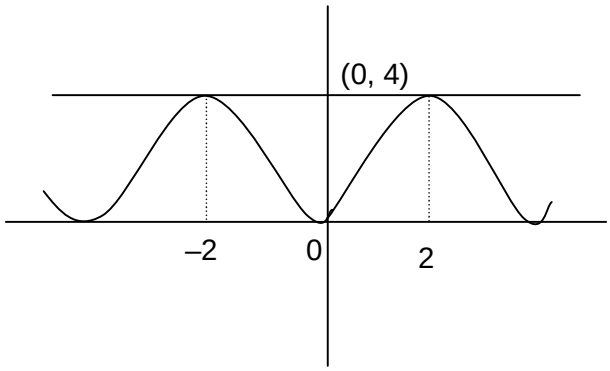
$$= \frac{1}{2}\ln 3 - \int_0^1 \frac{x^2-4+4}{x+2}dx - \frac{\ln 2}{2}$$

$$= \frac{1}{2}\ln\frac{3}{2} - \int_0^1\left((x-2) + \frac{4}{x+2}\right)dx \text{ now proceed}$$

84. B

Sol. Draw graphs of $y = x^2 . e^{2-|x|}$ and $y = 1$

$$y = \begin{cases} x^2.e^{2-x} & \text{if} \quad x \geq 0 \\ x^2.e^{2+x} & \text{if} \quad x < 0 \end{cases}$$



85. D

Sol. Differentiate $xy(x) = x^2 y'(x) + 2xy(x)$

or $xy(x) + x^2 y'(x) = 0$

$$x\frac{dy}{dx} + y = 0$$

$$\ln y + \ln x = \ln c$$

$$xy = c$$

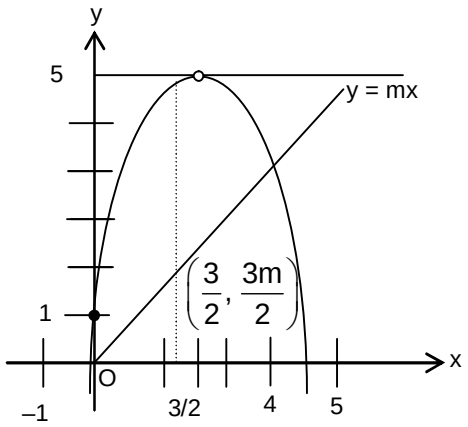
86. A

Sol. $$2.\frac{1}{2}\left(\frac{3}{2}.\frac{3m}{2}\right) = \int_0^{3/2} \left(1 + 4x - x^2\right) dx$$

$$\frac{9m}{4} = x + 2x^2 - \frac{x^3}{3}\Bigg|_0^{3/2}$$

$$= \frac{3}{2} + \frac{9}{2} - \frac{27}{8}.\frac{1}{3} = 6 - \frac{9}{8} = \frac{39}{8}$$

$$m = \frac{13}{6}$$



87. B

Sol. $N = 770 = 2 \times 5 \times 7 \times 11$

Hence number of ways $= 2^{n-1} = 2^3 = 8$

88. A

Sol. Vector $\left(3\hat{i} - 2\hat{j} + \hat{k}\right) \times \left(4\hat{i} - 3\hat{j} + 4\hat{k}\right)$ is perpendicular to $2\hat{i} - \hat{j} + m\hat{k}$.

$$\Rightarrow \begin{vmatrix} 3 & -2 & 1 \\ 4 & -3 & 4 \\ 2 & -1 & m \end{vmatrix} = 0$$

89. C

Sol. On (1, 2, 3) satisfies the plane $x-2y+z=0$ and also $\left(\hat{i}+2\hat{j}+3\hat{k}\right).\left(\hat{i}-2\hat{j}+\hat{k}\right)=0$

$\Rightarrow (a)$

Since the lines $\frac{x-1}{1}=\frac{y-2}{2}-\frac{z-3}{3}$ both satisfy $t+1, 2t+2$ and $3t+3$

Hence both are same $\Rightarrow (b)$.

Given lines is obviously $\|$ to the plane $x-2y+z=6 \Rightarrow (d)$

90. A

Sol. Given $(x+y)=(x+y)\left(x^2-xy+y^2\right)$

$\Rightarrow (x+y)\left(x^2-xy+y^2-1\right)=0$

$\therefore$ Either $x+y=0$ or $x^2-xy+y^2-1=0$

Now $x+y=0$ represent a line:

Also $\Delta=\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}=\begin{vmatrix} 1 & \frac{-1}{2} & 0 \\ \frac{-1}{2} & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix}=\frac{-3}{4}\neq 0$ and $h=\frac{-1}{2}, a=1=b,$ so $h^2<ab$

$\therefore$ $x^2-xy+y^2-1=0$ represent an ellipse.

Hence locus of $P(x,y)$ satisfying above equation is the union of line and an ellipse.